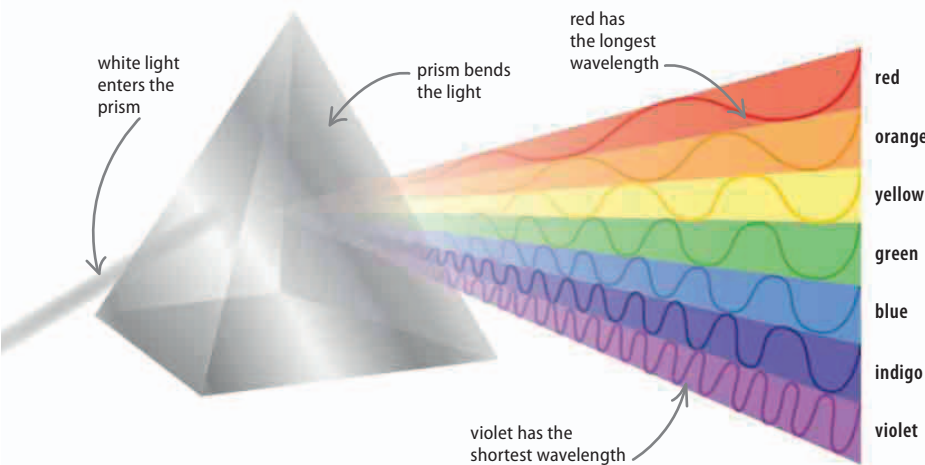


Light

LIGHT ENABLES US TO SEE A BRIGHT AND COLORFUL WORLD.

Light is the only type of electromagnetic radiation we can see. We are able to perceive it as a wide range of colors.



SEE ALSO

◀ 30–31 Photosynthesis

◀ 137 Light

◀ 194–195 Electromagnetic waves

Optics 198–199 ▶

Astronomy 230–231 ▶

Color spectrum

If white light is shined into a triangular block of glass, called a prism, the glass refracts (bends) the light. In an effect called dispersion, the light is split into different wavelengths, the band of visible colors known as the spectrum. The spectrum begins with the longest wavelength (red), and ends with the shortest wavelength (violet). Most people see seven distinct colors, but the spectrum is really continuous changing color.

◀ Splitting light

A prism bends light by different amounts, according to its wavelength.

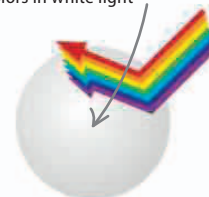
Making color

We see color based on information sent to the brain from millions of light-sensitive cells in the eye, called cones. There are three types of cone which respond to either red, green, or blue light. You see all colors as a mix of these three colors, known as primary colors.

▷ Reflective colors

Objects either reflect or absorb the different colors in white light. The reflected colors are the ones we see.

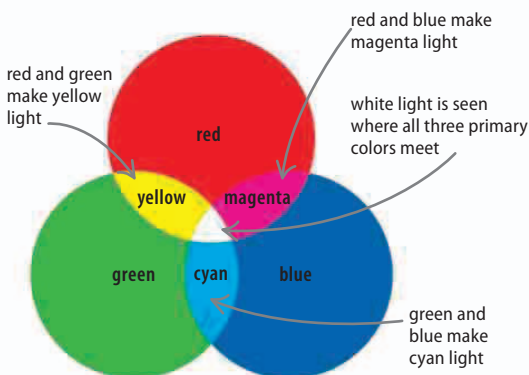
white objects reflect all colors in white light



yellow objects reflect yellow light and absorb other colors

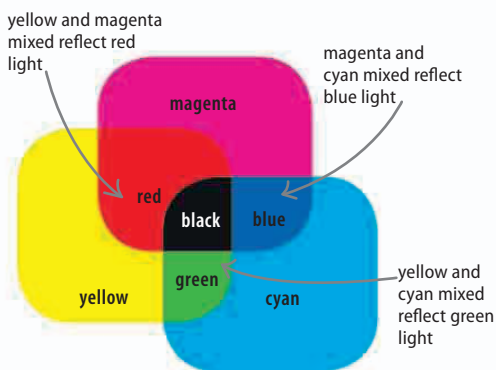


black objects absorb all colors and reflect none



△ Making colors with light

If you shine three flashlights, one of each of the primary colors, at a white surface, where they overlap they will create white light. Different combinations will create magenta, yellow, and cyan, known as secondary colors. This effect is used in televisions to create a full-color picture.



△ Mixing pigments

Making colors with pigments (inks and paints) is done in a very different way from colored light. The primary pigments are magenta, yellow, and cyan. Each reflects light of a different color. When the pigments are mixed the number of colors they reflect is reduced, and all three together make black.